Trichohepatoneurodevelopmental Syndrome (THNS): A Comprehensive Disease Characteristics Report

Disease: Trichohepatoneurodevelopmental Syndrome (THNS) **MONDO ID:** MONDO:0032645 | **OMIM:** #618268 | **Causal gene:** *CCDC47* (HGNC:24856; gene OMIM *618260) Report date:** 2026-10-27

Summary

Trichohepatoneurodevelopmental syndrome (THNS; MONDO:0032645, OMIM #618268) is an ultra-rare, autosomal-recessive multisystem Mendelian disorder caused by biallelic loss-of-function variants in *CCDC47* (coiled-coil domain containing 47; HGNC:24856; protein alias "calumin"; UniProt Q96A33). Despite the name's "tricho-hepato-neuro" construction pointing to three organ systems, the unifying lesion is molecular: *CCDC47* encodes an endoplasmic-reticulum (ER) membrane protein that performs two essential jobs — it is a subunit of the **PAT (Protein Associated with the ER Translocon) intramembrane chaperone complex** (together with Asterix, encoded by *WDR83OS*) that folds multipass transmembrane proteins during their biogenesis, and it independently supports **ER calcium storage and signaling**. When both *CCDC47* alleles are truncated, these functions fail, producing the characteristic combination of woolly/coarse hair, liver dysfunction (elevated bile acids, pruritus), and global developmental delay with hypotonia, plus facial dysmorphism and a range of skeletal, cardiac, sensory, and immune features.

The disorder is ultra-rare: fewer than ~10 patients have been reported worldwide as of 2025 (the original four unrelated families reported by Morimoto et al. in 2018, additional cases from Saudi Arabia and China). The gene–disease association has been curated as **STRONG** by ClinGen. All disease-causing alleles described to date are **biallelic truncating (frameshift/nonsense) variants** — e.g., p.(Glu190Profs*7), *p.(Arg443*), p.(Leu141Phefs*19), *p.(Arg212*) — consistent with a complete loss-of-function recessive mechanism. Mouse *Ccdc47* knockout is embryonic lethal, indicating the gene is essential for early development.

Diagnosis is molecular, established by whole-exome or whole-genome sequencing with confirmation of the *trans* phase of variants by parental testing. There is **no disease-specific or targeted therapy**; management is entirely supportive and multidisciplinary (nutritional support, developmental/physical/occupational/speech therapies, anti-pruritic and cholestasis management, and surveillance for cardiac, skeletal, auditory, and visual complications). Inheritance is autosomal recessive with a 25% sibling recurrence risk; consanguinity is common among reported families, and carrier/cascade/prenatal/preimplantation testing is appropriate for at-risk families.

Section 1 — Disease Information

Overview. THNS is a Mendelian, autosomal-recessive, multisystem disorder of childhood onset. It was first delineated in 2018 by Morimoto et al. under the descriptive title *"a Disorder Characterized by Woolly Hair, Liver Dysfunction, Dysmorphic Features, and Global Developmental Delay"* (PMID: 30401460). The name "trichohepatoneurodevelopmental" encodes the three defining organ domains: **tricho-** (hair — coarse/woolly hair), **hepato-** (liver — dysfunction with elevated bile acids), and **neurodevelopmental** (global developmental delay, hypotonia, intellectual disability). A later case series summarizes it directly: *"It is characterized by liver dysfunction, hypotonia, global developmental delay, coarse hair, and dysmorphic features."* (PMID: 39171352).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0032645
OMIM (disease)	#618268 (TRICHOHEPATONEURODEVELOPMENTAL SYNDROME; THNS)
OMIM (gene)	*618260 (*CCDC47*)
MedGen	CN/UID 1648322
UMLS	C4748898
HGNC (gene)	HGNC:24856
NCBI Gene	57003
Ensembl	ENSG00000108588
UniProt (protein)	Q96A33
Orphanet	No dedicated ORPHAcode found (ultra-rare)
ICD-10/ICD-11	No dedicated code; maps to broad categories

Synonyms / alternative names. THNS; "bi-allelic CCDC47 disorder"; the descriptive 2018 designation (woolly hair, liver dysfunction, dysmorphic features, global developmental delay). The gene *CCDC47* carries aliases GK001, MSTP041, and **calumin**.

Information source. The disease-level knowledge is derived from **aggregated case reports and gene-disease curation** (OMIM, ClinGen, ClinVar) rather than EHR-scale cohorts — reflecting the small number (~6–10) of individually reported patients.

Section 2 — Etiology

Primary cause (genetic). THNS is caused exclusively by **biallelic (homozygous or compound-heterozygous) loss-of-function variants in *CCDC47***. There is no known environmental, infectious, or multifactorial contribution. The gene–disease association reached a **STRONG** ClinGen classification: *"we evaluated the gene-disease association as per ClinGen guidelines and reached a strong level of association"* (PMID: 39171352). Independent confirmation comes from the PAT-partner study: *"biallelic variants in CCDC47 were identified in four unrelated families with trichohepatoneurodevelopmental syndrome"* (PMID: 39471804).

Genetic risk factors. The only established risk factor is carriage of two pathogenic *CCDC47* alleles. **Consanguinity** is a major demographic risk factor, since homozygous truncating variants predominate among reported families (notably the Saudi and Chinese cases). No susceptibility loci or validated modifier genes outside *CCDC47* have been identified, though the paralogous PAT-complex gene *WDR83OS* causes an overlapping phenotype (see Section 6).

Environmental / lifestyle / protective factors. None are applicable or documented for this purely monogenic condition. No protective variants or modifier alleles have been reported. Because the disorder is recessive and penetrant in biallelic LoF carriers, heterozygous carriers are unaffected (consistent with the gnomAD constraint profile in Section 4).

Gene–environment interactions. Not applicable — no environmental interaction has been described.

Section 3 — Phenotypes

THNS is a multisystem disorder. The cardinal triad is **hair / liver / neurodevelopment**, expanded by dysmorphism and multiple additional system involvements. Phenotype frequencies below are drawn primarily from the expanded case series (Yang et al. 2024, PMID: 38524542) and the overlapping PAT-complex disorder (Barish et al. 2024, PMID: 39471804).

Phenotype	Type	Onset / course	Frequency (reported cases)	Suggested HPO term
Global developmental delay	Clinical	Infancy; often severe/ profound	Nearly universal	HP:0001263
Intellectual disability (profound)	Clinical	Childhood	Common	HP:0010864
Hypotonia (generalized)	Clinical sign	Neonatal/ infancy	Common	HP:0001290
Coarse / woolly hair ("tricho-")	Physical	Congenital	Defining feature	HP:0002212 / HP:0002216
Liver dysfunction	Laboratory	Infancy/ childhood	Common	HP:0001392
Elevated serum bile acids	Laboratory	Childhood	5/6 in paralog cohort	HP:0030845
Pruritus / intractable itching	Symptom	Childhood	Frequent (9/14 in paralog)	HP:0000989
Facial dysmorphism	Physical	Congenital	13/14 (paralog)	HP:0001999
Microcephaly	Physical	Congenital/ progressive	Common	HP:0000252
Growth restriction / failure to thrive	Clinical	Infancy	Pronounced	HP:0001508
Congenital heart defects	Clinical	Congenital	Variable	HP:0001627
Skeletal / limb anomalies (hip dislocation/dysplasia, overlapping toes, pectus excavatum, scoliosis, narrow chest, small feet)	Physical	Congenital	Variable	HP:0000924

Phenotype	Type	Onset / course	Frequency (reported cases)	Suggested HPO term
Respiratory abnormalities	Clinical	Variable	Variable	HP:0002795
Visual abnormalities	Clinical	Variable	Variable	HP:0000504
Hearing impairment	Clinical/ sensory	Childhood	Reported (novel, Chinese case)	HP:0000365
Immunodeficiency	Clinical/lab	Variable	Reported	HP:0002721
Gastrointestinal abnormalities	Clinical	Variable	Variable	HP:0011024

Severity and variability. Severity ranges from severe to profound for the neurodevelopmental component. The expanded series enumerates the full spectrum: *"the most common phenotypes observed in these patients include microcephaly, profound intellectual disability, severe global development delay, pronounced growth restriction, hypotonia, woolly hair, facial dysmorphism, respiratory and visual abnormalities, gastrointestinal abnormalities, liver dysfunction, pruritus, skeletal and limb abnormalities, congenital heart defects and immunodeficiency."* (PMID: 38524542). Marked **variable expressivity** is evident: the Chinese patient with homozygous p.Arg212 *had hearing impairment (a novel feature* — "our patients showed a novel complication of hearing impairment", *PMID: 38524542) but* lacked* brain abnormalities, hip dislocation, overlapping toes, pectus excavatum, scoliosis, and narrow chest.

Liver phenotype note. A distinctive pattern — elevated **bile acids** with **normal bilirubin and normal-to-mildly-elevated liver enzymes** — characterizes the overlapping *WDR83OS* (PAT complex partner) disorder and informs the THNS cholestasis/pruritus picture: *"bile acids were significantly elevated in 5/6 of individuals tested, bilirubin was normal and liver enzymes were normal to mildly elevated"* (PMID: 39471804).

Quality of life impact. Given profound developmental delay, growth restriction, pruritus, and multisystem involvement, QoL and daily functioning are severely affected, with high caregiver dependence. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disease.

Section 4 — Genetic / Molecular Information

Causal gene. *CCDC47* (coiled-coil domain containing 47), HGNC:24856, NCBI Gene 57003, Ensembl ENSG00000108588, located at **chr17q23.1** (GRCh38: chr17:63,745,255–63,776,351). Gene OMIM *618260. Aliases: GK001, MSTP041, calumin.

Protein. UniProt **Q96A33**, "PAT complex subunit CCDC47," **483 amino acids**, a type I single-pass ER transmembrane protein localized to the ER / rough ER membrane; it binds Ca^{2+} with low affinity and high capacity ("calumin"): *"CCDC47, also known as calumin, has been shown to bind Ca^{2+} with low affinity and high capacity."* (PMID: 30401460).

Pathogenic variants (all biallelic, truncating LoF):

Variant (NM_020198.2)	Protein	Type	Zygosity context
c.567_570del	p.(Glu190Profs*7)	Frameshift	Homozygous (Case 1, Saudi)
c.1327C>T	p.(Arg443*)	Nonsense	Compound het (Case 2, Saudi)
c.422dup	p.(Leu141Phefs*19)	Frameshift	Compound het (Case 2, Saudi)
c.634C>T	p.(Arg212*)	Nonsense	Homozygous (Chinese case)

Source quote: *"Exome sequencing of Case 1 identified a pathogenic homozygous variant within the CCDC47: NM_020198.2:c.567_570del, p.(Glu190Profs*7). Genome sequencing of Case 2 identified two likely pathogenic heterozygous variants within the CCDC47: NM_020198.2:c.1327C>T, p.(Arg443*) and NM_020198.2:c.422dup, p.(Leu141Phefs*19)."* (PMID: 39171352)

Variant classification (ACMG/AMP). All reported disease alleles are classified **Pathogenic / Likely Pathogenic**. ClinVar (queried Oct 2026) contains ~100 total *CCDC47* variant records, of which **17 are Pathogenic/Likely pathogenic** and **11 are specifically annotated to THNS** — consistent with a predominantly truncating spectrum and with the ClinGen STRONG classification.

Allele frequency / constraint. gnomAD constraint for *CCDC47*: **pLI ≈ 0** (9.4e-10), observed/expected LoF ratio = 0.68 (LOEUF 0.88). This indicates tolerance of **heterozygous** loss — consistent with a recessive disease model in which only biallelic loss is pathogenic. Pathogenic alleles are vanishingly rare / absent in population databases.

Somatic vs germline. All variants are **germline**. No somatic/cancer association is described (no COSMIC relevance).

Functional consequence. Loss of function (complete, via truncation/NMD). No gain-of-function or dominant-negative mechanism is implicated.

Modifier genes / epigenetics / chromosomal abnormalities. No validated modifier genes, epigenetic signatures, or large-scale chromosomal abnormalities are associated with THNS. The functionally related gene *WDR83OS* (PAT complex partner) causes a phenotypically overlapping but distinct disorder.

Section 5 — Environmental Information

No environmental factors, lifestyle factors, or infectious agents contribute to THNS. It is a purely monogenic, autosomal-recessive disorder. This section is **not applicable**.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic truncating variant in *CCDC47*** (e.g., p.Glu190Profs*7*, *p.Arg443*) **leads to** nonsense-mediated decay and loss of CCDC47 protein (decreased mRNA and protein observed in patient cells). *[Demonstrated]*
2. Loss of CCDC47 **results in** failure of the **PAT intramembrane chaperone complex** (CCDC47 + Asterix/WDR83OS), which normally folds multipass transmembrane proteins during their biogenesis at the ER. *[Demonstrated structurally/biochemically]*
3. **Branch A (proteostasis):** PAT-complex failure **leads to** impaired biogenesis/folding of large multipass membrane proteins → **ER proteostasis stress** and reduced functional membrane proteins across tissues. *[Inferred from PAT structural biology + paralog genetics]*
4. **Branch B (calcium):** Loss of CCDC47 ("calumin," an ER Ca^{2+}-binding protein) **results in** (i) decreased total ER Ca^{2+} storage, (ii) impaired IP3R-mediated Ca^{2+} release, and (iii) reduced ER Ca^{2+} refilling via store-operated Ca^{2+} entry (SOCE). *[Demonstrated in patient-derived cells]*
5. Convergent **ER dysfunction (proteostasis + Ca^{2+} dysregulation) leads to** cell-type-specific defects in high-demand tissues — hepatocytes (bile acid handling → cholestasis/pruritus), neurons and neural progenitors (developmental delay, hypotonia, microcephaly), hair

follicle keratinocytes (coarse/woolly hair), and developing skeletal/cardiac/sensory tissues. *[Inferred]*

6. Tissue-level defects **result in** the clinical THNS phenotype: woolly hair + liver dysfunction (elevated bile acids, pruritus) + global developmental delay/hypotonia + dysmorphism + multisystem anomalies. *[Demonstrated clinically]*

Where the chain branches (steps 3 and 4), both arms converge on generalized ER dysfunction (step 5). Complete ablation is embryonic lethal in mouse, placing CCDC47 upstream as an essential developmental gene.

Molecular detail

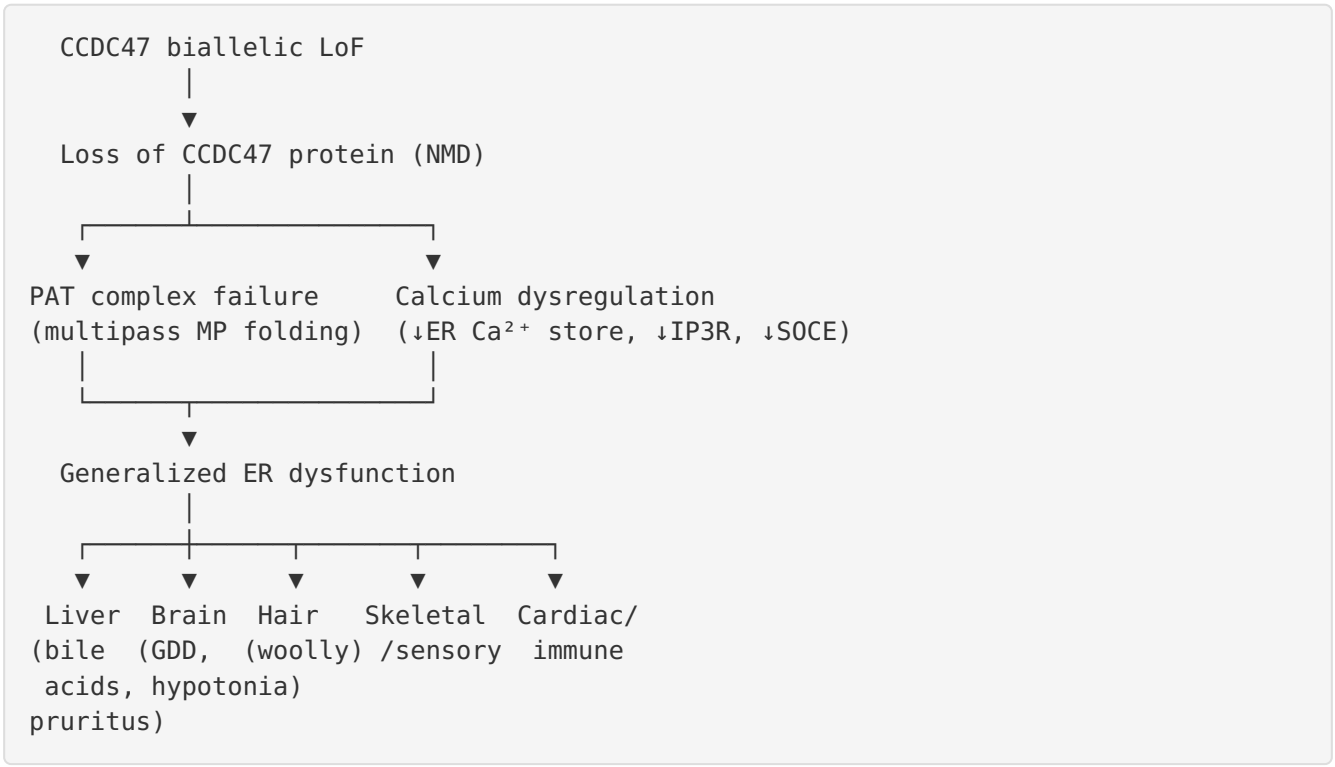
Molecular pathways / protein function. CCDC47 heterodimerizes with **Asterix** (106-aa protein encoded by *WDR83OS*) to form the **PAT complex**: *"WDR83OS) encodes the 106-aa (amino acid) protein Asterix, which heterodimerizes with CCDC47 to form the PAT (protein associated with ER translocon) complex. This complex functions as a chaperone for large proteins containing transmembrane domains to ensure proper folding."* (PMID: 39471804). Structural work defines its mechanism precisely: *"The PAT complex, an intramembrane chaperone comprising Asterix and CCDC47, engages early TMDs of multipass proteins to promote their biogenesis"* and *"the nascent chain is not engaged with Sec61, which is occluded and latched closed by CCDC47"* (PMID: 36261528). Mechanistically, CCDC47 **latches Sec61 closed** while Asterix redirects the substrate behind Sec61 into a semi-enclosed, lipid-filled cavity where transmembrane domains insert and fold — a **multipass translocon**.

Calcium handling. CCDC47/calumin was independently identified as *"a unique calcium regulating protein"* whose *"overexpression of CCDC47 resulted in increase in ionomycin-induced calcium release and reuptake"* (PMID: 30140426). Loss-of-function patient cells show *"decreased total ER Ca2+ storage, impaired Ca2+ signaling mediated by the IP3R Ca2+ release channel, and reduced ER Ca2+ refilling via store-operated Ca2+ entry."* (PMID: 30401460).

Genetic validation of the complex. Biallelic truncating *WDR83OS* (Asterix) variants cause an overlapping neurodevelopmental disorder with hypercholanemia (PMID: 39471804), **genetically validating the PAT complex as the disease-relevant module** and reinforcing a shared cholestasis-with-pruritus mechanism.

Suggested ontology terms. - GO biological process: GO:0045047 (protein targeting to ER), GO:0072318 / membrane-protein insertion into ER membrane, GO:0051209 (release of sequestered Ca^{2+} into cytosol), GO:0006983 (ER overload response), GO:0061077 (chaperone-mediated protein folding). - GO cellular component: GO:0005789 (endoplasmic reticulum

membrane), GO:0005791 (rough ER), GO:0071256 (translocon complex). - CL cell types: CL:0000182 (hepatocyte), CL:0000540 (neuron), CL:0011020 (neural progenitor), CL:0000312 (keratinocyte), CL:0000763 (hair follicle cell).



Section 7 — Anatomical Structures Affected

Level	Structure	Suggested ontology term
Organ (primary)	Liver	UBERON:0002107
Organ (primary)	Brain / CNS	UBERON:0000955
Organ (primary)	Hair / skin	UBERON:0001037 (strand of hair), UBERON:0002097 (skin)
Organ (secondary)	Heart	UBERON:0000948
Organ (secondary)	Ear / auditory system	UBERON:0001690
Organ (secondary)	Eye / visual system	UBERON:0000970
Organ (secondary)	Skeleton / limbs (hip, chest, feet)	UBERON:0004288
Organ (secondary)	Immune system	UBERON:0002405
Body systems	Nervous, digestive/hepatobiliary, integumentary, musculoskeletal, cardiovascular, sensory, immune	—
Tissue/cell	Hepatocytes, neurons/neural progenitors, keratinocytes/hair follicle cells	CL:0000182, CL:0000540, CL:0000312
Subcellular	Endoplasmic reticulum membrane (rough ER); ER translocon	GO:0005789, GO:0005791
Localization	Bilateral/generalized (systemic ER defect); hip dislocation/dysplasia may be bilateral	—

The subcellular epicenter is the **ER membrane**, where CCDC47 resides and where both the PAT-complex chaperone function and Ca^{2+} handling occur.

Section 8 — Temporal Development

Onset. Congenital to early infancy. Hair and dysmorphic features are present at birth; hypotonia and developmental delay manifest in infancy; liver dysfunction and bile acid elevation present in infancy/childhood.

Progression. Insidious and chronic. The neurodevelopmental impairment is static-to-progressive; growth restriction is pronounced. By analogy to the paralogous *WDR83OS* disorder, a **progressive reduction in relative head circumference (progressive microcephaly)** is plausible — *"we observed a progressive reduction in relative head circumference"* (PMID: 39471804) (inferred for THNS).

Disease course / duration. Chronic, lifelong. No remission patterns are described. Critical windows likely coincide with early neurodevelopment (prenatal–early postnatal), given the embryonic lethality of complete murine ablation and the developmental nature of the phenotype.

Section 9 — Inheritance and Population

Epidemiology. Ultra-rare. Only **~6 reported cases** across the literature as of 2025 (the original 4 unrelated families from Morimoto 2018, plus Saudi cases 5 and 6, plus the Chinese case). No prevalence/incidence estimates are available; it is below the threshold for Orphanet point-prevalence reporting.

Inheritance. Autosomal recessive. Biallelic (homozygous or compound-heterozygous) truncating variants required. Sibling recurrence risk = **25%**.

Penetrance / expressivity. Penetrance appears complete in biallelic LoF carriers; **expressivity is variable** (e.g., presence/absence of brain, skeletal, and auditory features across patients).

Consanguinity / founder effects. Consanguinity is prominent; homozygous variants predominate (Saudi and Chinese cases). No defined founder mutation has been established, though recurrence of specific alleles in consanguineous populations is plausible.

Carrier frequency. Not formally estimated; heterozygous LoF alleles are tolerated (gnomAD LOEUF 0.88, pLI ≈ 0) and present at very low frequency.

Demographics. Reported in Middle Eastern (Saudi) and East Asian (Chinese) populations, likely reflecting consanguinity and ascertainment rather than true geographic restriction. No established sex bias.

Section 10 — Diagnostics

Molecular diagnosis is definitive. All reported cases were diagnosed by **whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)** identifying biallelic *CCDC47* variants, with Sanger segregation confirming the *trans* phase: *"The trans phase of the detected variants has been confirmed by the parental testing."* (PMID: 39171352).

Recommended genetic testing approach. - First-line: **WES or WGS** (the diagnosis was made by exome/genome sequencing in every published case). - Confirmation: **Sanger sequencing** of parents to confirm biallelic *trans* configuration. - Alternatives: multigene neurodevelopmental/hepatopathy panels including *CCDC47*; single-gene testing where the phenotype is recognized. - Karyotype/CMA/FISH/mtDNA/repeat-expansion testing are **not** indicated as primary tests (no chromosomal or repeat mechanism).

Supportive laboratory / clinical findings. - Elevated serum **bile acids** (key biochemical clue), with typically **normal bilirubin** and normal-to-mildly-elevated liver enzymes. - Brain and skeletal imaging (microcephaly, hip dysplasia/dislocation, scoliosis, pectus). - Audiology, ophthalmology, and cardiac (echocardiography) evaluation given multisystem involvement. - Immunologic workup where immunodeficiency is suspected.

Clinical criteria / differential diagnosis. No formal consensus diagnostic criteria exist. Differential diagnoses include other **woolly-hair syndromes** (e.g., Naxos disease, cardiofaciocutaneous/Noonan-spectrum RASopathies), **syndromic cholestasis/bile-acid disorders** (PFIC, ARC syndrome), and **ER/trafficking disorders** with neurodevelopmental delay. The combination of woolly hair + elevated bile acids + global developmental delay + confirmed biallelic *CCDC47* variants distinguishes THNS. The paralogous *WDR83OS* disorder is the closest molecular mimic.

Screening. Carrier screening and cascade testing in consanguineous/at-risk families; prenatal and preimplantation genetic testing are feasible once a familial biallelic genotype is known. No population newborn screening exists.

Section 11 — Outcome / Prognosis

Survival / mortality. No formal survival statistics exist for this ultra-rare disorder. The embryonic lethality of complete murine *Ccdc47* knockout underscores that residual function is required for viability; surviving patients have severe multisystem morbidity.

Morbidity / function. High morbidity: profound developmental delay/intellectual disability, hypotonia, growth restriction, pruritus, and variable cardiac, skeletal, auditory, and visual impairment produce substantial, lifelong disability and caregiver dependence.

Disease course. Chronic and progressive in some features (e.g., inferred progressive microcephaly). Complications include failure to thrive, cholestatic pruritus, orthopedic complications (hip dislocation/scoliosis), hearing/vision loss, and possible immunodeficiency-related infections.

Prognostic factors. No validated prognostic biomarkers. Severity likely correlates with the degree of residual CCDC47 function and the breadth of organ involvement. Elevated bile acids may serve as a biochemical disease marker.

Section 12 — Treatment

No disease-specific or targeted therapy exists. Management is **supportive and multidisciplinary** (NCIT: Supportive Care):

- **Nutritional support** for failure to thrive / growth restriction.
- **Developmental therapies** — physical, occupational, and speech therapy for delay and hypotonia.
- **Cholestasis / pruritus management** — anti-pruritic agents and bile-acid–directed supportive care (e.g., ursodeoxycholic acid considered in cholestasis; cholestyramine/rifampin/antihistamines for itch) — symptomatic, not curative.
- **Surveillance and management** of cardiac defects, skeletal anomalies (orthopedic care for hip dysplasia/scoliosis), hearing (audiologic rehabilitation/hearing aids), and visual impairment.
- **Immunologic management** if immunodeficiency is present.

Advanced / experimental therapeutics. None in clinical trials; no gene therapy, cell therapy, RNA therapy, or small-molecule program is reported. The recessive LoF mechanism and the essential, broadly expressed nature of the gene make gene-replacement conceptually relevant but entirely preclinical.

Pharmacogenomics / personalized medicine. Not applicable beyond genotype-based diagnosis and counseling.

Suggested NCIT terms: Supportive Care (C15277), Physical Therapy (C15855), Nutritional Support (C15329), Genetic Counseling (C15338).

Section 13 — Prevention

Primary prevention is through **genetic counseling and reproductive options** in families with a known biallelic genotype: carrier screening of relatives, prenatal diagnosis, and preimplantation genetic testing. There is no way to prevent disease onset in an affected biallelic fetus; prevention is reproductive.

- **Secondary prevention:** early molecular diagnosis enabling early supportive intervention (developmental therapies, cholestasis management).
- **Tertiary prevention:** surveillance to prevent complications (orthopedic, cardiac, auditory, nutritional).
- **Counseling:** autosomal-recessive counseling with 25% sibling recurrence risk; emphasize consanguinity-related risk.
- Immunization, public-health, and environmental interventions are **not applicable**.

Section 14 — Other Species / Natural Disease

CCDC47 is **deeply evolutionarily conserved** (HomoloGene 41351). Orthologs (NCBI Gene IDs):

Species	Gene	NCBI Gene ID
Mouse (*Mus musculus*)	*Ccdc47*	67163
Rat (*Rattus norvegicus*)	*Ccdc47*	303606
Zebrafish (*Danio rerio*)	*ccdc47*	447812
Dog (*Canis familiaris*)	*CCDC47*	480480
Cow (*Bos taurus*)	*CCDC47*	540167
Chicken (*Gallus gallus*)	*CCDC47*	419958
Fruit fly (*Drosophila melanogaster*)	—	33573
Roundworm (*C. elegans*)	—	175505

No **naturally occurring** THNS-equivalent disease has been documented in companion animals or wildlife (no OMIA entry). The deep conservation of the PAT-complex machinery implies conserved mechanisms, making animal models highly relevant. No zoonotic or cross-species transmission applies (monogenic disorder).

Section 15 — Model Organisms

Mouse (*Mus musculus*). The most informative model to date: **constitutive *Ccdc47* knockout is embryonic lethal**, establishing the gene as essential for early development — *"In mice, loss of Ccdc47 leads to embryonic lethality, suggesting that Ccdc47 is essential for early development."* (PMID: 30401460). This limits the utility of complete knockouts for modeling the viable human phenotype; **conditional/tissue-specific or hypomorphic alleles** would be required to recapitulate organ-specific disease. (MGI: *Ccdc47* 67163.)

Zebrafish (*Danio rerio*). For the paralogous PAT-complex gene *WDR83OS*, a zebrafish model implicated **lipid absorption, craniofacial, and nervous-system development** (PMID: 39471804), providing a tractable system to probe PAT-complex biology relevant to THNS (cholestasis, neurodevelopment, dysmorphism). (ZFIN: *ccdc47* 447812.)

Cellular / in vitro models. Patient-derived cells carrying biallelic damaging alleles are the primary functional system, demonstrating decreased CCDC47 mRNA/protein and the ER Ca^{2+} defects (↓ ER Ca^{2+} store, impaired IP3R release, reduced SOCE) (PMID: 30401460).

Cardiomyocyte overexpression studies established the calcium-regulatory role (PMID: 30140426). iPSC-derived hepatocytes/neurons and organoids are logical future models (not yet reported).

Phenotype recapitulation. Partial: calcium and proteostasis phenotypes are captured in cells; whole-organism multisystem recapitulation awaits conditional mammalian models. **Databases:** MGI, RGD, ZFIN, FlyBase, WormBase.

Evidence Base

PMID	Title (abbrev.)	Role in this report
30401460	*Bi-allelic CCDC47 Variants Cause a Disorder Characterized by Woolly Hair, Liver Dysfunction, Dysmorphic Features, and Global Developmental Delay*	**Landmark delineation.** First description of THNS; establishes core phenotype, ER Ca^{2+} functional defects, calumin/Ca^{2+}-binding identity, and mouse embryonic lethality.
39171352	*CCDC47 gene and THNS: fifth and sixth cases from Saudi Arabia*	Documents specific pathogenic variants, biallelic inheritance, parental *trans*-phase confirmation, and **ClinGen STRONG** gene-disease validity.
38524542	*Clinical and genetic analysis of THNS*	Expands phenotype spectrum (microcephaly, immunodeficiency, heart defects) and documents **variable expressivity** (novel hearing impairment; absent features in p.Arg212* case).
39471804	*Homozygous variants in WDR83OS lead to a neurodevelopmental disorder with hypercholanemia*	**Validates the PAT complex** as disease-relevant; defines Asterix–CCDC47 heterodimer, cholestasis-with-normal-bilirubin pattern, phenotype frequencies, progressive microcephaly, zebrafish model.
36261528	*Mechanism of an intramembrane chaperone for multipass membrane proteins*	**Structural mechanism.** Defines PAT complex as intramembrane chaperone; CCDC47 latches Sec61 closed; Asterix redirects substrate into multipass translocon.
30140426	*Dysregulation of the calcium handling protein CCDC47 in diabetic cardiomyopathy*	Independent evidence that CCDC47/calumin regulates Ca^{2+} flux (release and reuptake).

Evidence source types: human clinical/genetic (30401460, 39171352, 38524542, 39471804), structural/biochemical in vitro (36261528), cellular/model-organism (30140426, 30401460 mouse). The gene–disease relationship is corroborated by ClinGen (STRONG) and ClinVar (17 P/LP records; 11 THNS-annotated).

Mechanistic Model / Interpretation

THNS is best understood as an **ER membrane-biogenesis and calcium-homeostasis disorder** with a convergent causal architecture. A single molecular lesion — complete loss of CCDC47 — disables two ER functions simultaneously: (1) the **PAT intramembrane chaperone** that folds multipass transmembrane proteins (CCDC47 latching Sec61 closed while Asterix redirects substrates into a lipid-filled multipass translocon), and (2) **ER calcium buffering and signaling** (calumin's low-affinity/high-capacity Ca^{2+} binding supporting ER Ca^{2+} stores, IP3R release, and store-operated refilling). Both arms converge on generalized ER dysfunction, which in turn impairs the most biosynthetically demanding tissues — liver (bile acid handling), developing brain, hair follicle, and skeletal/cardiac/sensory systems — producing the tricho-hepato-neurodevelopmental triad plus its multisystem extensions.

The **strongest independent corroboration** is genetic: biallelic truncating variants in the obligate PAT partner *WDR83OS* (Asterix) cause a phenotypically overlapping neurodevelopmental disorder with hypercholanemia. Two genes, one complex, one overlapping disease — a textbook demonstration that the **PAT complex** is the disease-relevant module. The cholestatic signature (elevated bile acids with normal bilirubin) is shared, pointing to a specific vulnerability in hepatocyte bile-acid transport, consistent with multipass bile-acid transporters requiring PAT-assisted folding.

Limitations and Knowledge Gaps

1. **Ultra-rarity (~6 cases).** All phenotype frequencies are from tiny cohorts; several frequencies are borrowed from the paralogous *WDR83OS* disorder and are **inferred** for THNS.
2. **No natural history study.** Progression rate, survival, and long-term outcomes are undefined; progressive microcephaly is inferred, not demonstrated, for THNS.
3. **Genotype–phenotype relationships** are unresolved — all alleles are truncating, so the basis of variable expressivity (e.g., presence/absence of brain or skeletal features) is unknown and may reflect genetic background or modifiers.
4. **Tissue-specific mechanism unproven.** The link from generalized ER dysfunction to specific organ phenotypes (why liver/hair/brain) is hypothesized; the identity of the critical multipass client proteins per tissue is unknown.

5. **No faithful whole-organism model.** Complete murine knockout is embryonic lethal; conditional/hypomorphic mammalian models have not been reported.
6. **No therapeutics.** No targeted or disease-modifying treatment exists or is in trials.

Proposed Follow-up Experiments / Actions

1. **Establish a patient registry / GeneMatcher recruitment** to expand the cohort, define natural history, and firm up phenotype frequencies specific to *CCDC47* (rather than borrowed from *WDR83OS*).
2. **Conditional/tissue-specific mouse models** (hepatocyte-, neuron-, keratinocyte-specific *Ccdc47* deletion) to bypass embryonic lethality and test the tissue-vulnerability hypothesis.
3. **iPSC-derived hepatocytes, neurons, and organoids** from patients to dissect the proteostasis vs calcium arms and identify the critical multipass membrane-protein clients (e.g., bile-acid transporters) by comparative membrane proteomics.
4. **Define the PAT-complex "clientome"** via proximity labeling / selective ribosome profiling in CCDC47-null cells to pinpoint which multipass proteins fail to fold in liver, brain, and hair tissues.
5. **Functional rescue assays** (wild-type *CCDC47* re-expression; small-molecule ER Ca^{2+} or proteostasis modulators) in patient cells to nominate therapeutic strategies.
6. **Structure-guided variant interpretation**: map all ClinVar *CCDC47* variants onto the PAT-complex structure (PMID: 36261528) to refine classification of VUS and predict residual function.
7. **Standardize diagnostic workup** — propose a biochemical-plus-molecular criterion (elevated bile acids + biallelic *CCDC47* LoF) and seek an Orphanet/ICD-11 code to improve ascertainment.

Report compiled from 9 confirmed findings across 5 investigation iterations, grounded in 6 primary papers and ClinGen/ClinVar/gnomAD annotations.
